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import java.util.HashMap;
import java.util.Scanner;

class User {
	private String username;
	private String password;
	public double balance;
	public double loanAmount;
	private String pinCode;

	public User(String username, String password, String pinCode, double balance) {
		this.username = username;
		this.password = password;
		this.pinCode = pinCode;
		this.balance = balance;
		this.loanAmount = 0;
	}
	public boolean authenticate(String inputUser, String inputPass) {
		return this.username.equals(inputUser) && this.password.equals(inputPass);
	}
	public boolean validatePin(String inputPin) {
		return this.pinCode.equals(inputPin);
	}
	public void deposit(double amount) {
		if (amount <= balance) {
			balance -= amount;
			return true;
		}
		return false;
	}
	public void applyLoan(double amount) {
		loanAmount += amount;
		balance += amount;
	}
	public boolean  makeLoanPayment(double amount) {
		if (amount <= balance && amount <= loanAmount) {
			loanAmount -= amount;
			balance -= amount;
			return true;
		}
		return false;
	}
	public double getBalance() {
		return balance;
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	}
	public String getDetails() {
		return "User: " + username + "\nBalance: ₱" + loanAmount;
	}
}
public class BankSystemConsole {
	public static HashMap<String, User> users = new HashMap<>();
	private String loanDetails = "No loan Application subbmitted.";
	private static Scanner scanner = new scanner(System.in);

	public static void main(String[] arg) {
		users.put("victorino", new User("victorino", "password123", "1234", 7000));

		System.out.print("Enter username: ");
		String username = scanner.nextLine();
		System.out.print("Enter password: ");
		String password = scanner.nextLine();
		if (user.contaisKey(username) && users.get(username).authenticate(username,
password)) {
			User currentUser = users.get(username);
			System.out.println("Welcome, " + username + "!");
			new BankSystemConsole().processTransactions(currentUser);
		} else {
			System.out.println("Invalid login credentials.");
		}
	}
	private void processTransactions(User user) {
		int option = 0;
		do {
			System.out.println("\nMenu:");
			System.out.println("1. Deposit");
			System.out.println("2. Withdrawal");
			System.out.println("3. Apply Loan");
			System.out.println("4. Make Loan Payment");
			System.out.println("5. View Balance");
			System.out.println("6. View Loan Detwils");
			System.out.println("7. Exit");
			System.out.print("Choose Option: ");

			try {
				option = Integer.parseInt(scanner.nextLine());
			} catch (NumberFormatException e) {
				System.out.println("Invalid input! Please enter a valid number.");
				continue;
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                        }
                        switch (option) {
                        case 1:
                                System.out.print("Enter Deposit Amount: ");
                                double depositAmount = Double.parseDouble(scanner.nextLine());
                                user.deposit(depositAmount);
                                System.out.println("Deposit successful! Your Balance is: ₱" +
user.getBalance());
                                break;
                        case 2:
                                handleWithdrawal(user);
                                break;
                        case 3:
                                handleLoanApplication(user);
                                break;
                        case 4:
                                System.out.print("Enter Loan payment amount: ");
                                double paymentAmount =
Double.parseDouble(scanner.nextLine());
                                if (user.makeLoanPayment(paymentAmount)) {
                                        System.out.println("Loan Payment successful! Your update
balance is: ₱" + user.getBalance());
                                } else {
                                        System.out.println("Insufficient funds or invalid amount");
                                }
                                break;
                        case 5:
                                System.out.println(user.getDetails());
                                break;
                        case 6:
                        System.out.println(loanDetails);
                        break;
                        case 7:
                        System.out.println("Thank you for Banking with us");
                        break;
                        default:
                        System.out.println("Invalid option. Try Again.");
                        }
                } while (option != 7);
        }
        private void handleWidrawal(User user) {
                System.out.println("Withdrawal form:");
                System.out.println("1. Savings");
                System.out.println("2. Credit");
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                Syatem.out.print("Enter Choice: ");

                int accountChoice = Integer.parseInt(Scanner.nextLine());
                if (accountChoice != 1 && accountChoice != 2) {
                        System.out.println("Invalid account type.");
                        return;
                }
                System.out.print("Enter PIN to proceed: ");
                String pinInput = scanner.nextLine();
                if (!user.validatePin(pinInput)) {
                        System.out.println("Invalid PIN.");
                        return;
                }
                System.out.print("Enter Withdrawal amount: ");
                double withdrawalAmount = Double.parseDouble(scanner.nextLine());
                if (user.withdrawal(withdrawAmount)) {
                        System.out.println("Withdrawal succcessful! Your remaining balance is: ₱"
+ user.getBalance());
                } else {
                        System.out.println("Insufficient funds."); }
        }
        private void handleLoanApplication(User user) {
                System.out.print("Enter full name: ");
                String name = scanner.nextLine();
                System.out.print("Enter work Type (1. Goverment , 2. Private, 3. Own
Business):");
                String workType = scanner.nextLine();
                System.out.print("Enter contact number: ");
                String contactNumber = scanner.nextLine();
                System.out.print("Enter email address: ");
                String email = scanner.nextLine();
                System.out.print("Enter your full Address: ");
                String address = scanner.nextLine();

                System.out.println("Select ID to submit:");
                System.out.println("1. Driver's License \n2. TIN ID \n3. Goverment ID \n4.
National ID");
                System.out.print("Enter choice: ");
                String selectedID = scanner.nextLine();

                System.out.print("Enter loan amount (Whole number only): ");
                int laonAmount = Integer.parseInt(scanner.nextLine());

                System.out.println("Select loan teem:");
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                System.out.println("1. 6 months - 3% per month");
                System.out.println("2. 12 months - 5% per month");
                System.out.println("3. 15 months - 6% per month");
                System.out.println("4. 24 months - 8% per month");
                System.out.print("Enter choice: ");
                int months = Integer.parseInt(scanner.nextLine().split(" ")[0]);

                int monthlyPayment = loanAmount / months;

                System.out.println("\nLoan Details: ");
                System.out.println("Loan Amount: ₱" + loanAmount);
                System.out.println("Monthly Payment: ₱" + monthlyPayment);
                System.out.println("Do you agree to the loan terms? (yes/no):");
                String agreement = scanner.nextLine();

                if (agreement.equalsIgnoreCase("yes")) {
                        loanDetails = "Loan submitted! \nApplication Name: " + name + "\nWork
Type: " + workType + "\nContact: " + contactNumber + "\nEmail: " + email + "\nAddress: " +
address + "\nSubmitted ID: " + selectedID + "\nLoan amount: " + "\nMonthly Payment: ₱" +
monthlyPayment;

                        System.out.println("Loan Application submitted successfully!");
                } else {
                        System.out.println("Loan Application cancelled.");
                }
        }
}
```